

Construct-Specified LLM Measurement for Text Artefacts

From holistic prediction to validity-guided measurement

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External thesis project with Barter

Second-reader overview

June 23, 2026

Motivation

The challenges of artefact-level evaluation

- LLMs are increasingly used to produce open-ended text in operational settings.
- Evaluation is therefore not only about comparing systems, but also about assessing individual outputs.
- System-level benchmarks ask: *How capable is the model?*
- Artefact-level diagnostics ask: *What is good or bad about this specific text?*
- **Problem:** “good” is not self-evident; it depends on which dimensions are taken to constitute quality.

Thesis motivation

Artefact-level evaluation requires explicit construct specification: defining quality dimensions, measuring them, and validating whether they support the intended quality claim.

Literature lineage

Why artefact-level LLM measurement matters

- Classic evaluation methods were limited to surface-level overlap [1, 2] or semantic similarity [3, 4]
- LLMs solved these problems by acting as **general-purpose evaluators** [5, 6]
- Current frontier: automatically evaluating text on multiple dimensions [7–9]
 - This adheres to the multi-dimensional nature of text quality [10]

Core problem

Multi-dimensional LLM evaluation produces criterion-level scores, but it is often unclear how these scores should be recomposed into a meaningful claim about artefact quality.

Thesis response

Define artefact evaluation as a measurement problem: quality dimensions are theoretically specified, measured separately with LLMs, and recomposed to collect validity evidence for the resulting construct-level measurement claim.

Measurement framework

Psychometric basis

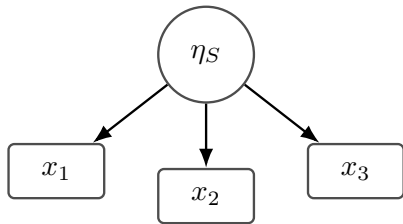
The measurement problem: three requirements

- ① **Target construct:** what is being measured?
 - ② **Measurement model:** how do scores relate to the construct?
 - ③ **Validity evidence:** what evidence supports the interpretation?
- Psychometrics distinguishes between **reflective** and **formative** measurement models [11].
 - This distinction determines how criteria relate to the target construct, what the main validity risks are, and what kind of evidence is needed.

Measurement framework

Measurement models

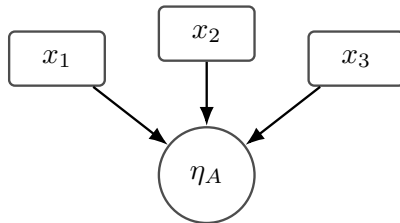
System-level evaluation Reflective measurement



Latent system ability η_S causes observed performance across tasks, prompts, or benchmark items.

$$x_j = \lambda_j \eta_S + \varepsilon_j$$

Artefact-level evaluation Formative measurement



Criterion-specific scores jointly constitute the quality η_A of an individual artefact.

$$\eta_A = \sum_{j=1}^J \gamma_j x_j + \zeta$$

Measurement framework

Implication for the empirical design

- In formative measurement, the criterion set is part of the construct specification: changing the criteria changes the construct [11]
- Validity evidence is collected by relating LLM-derived measurements to external validation criteria
- The empirical question is whether explicit formative specification yields stronger validity evidence and more interpretable diagnostics.

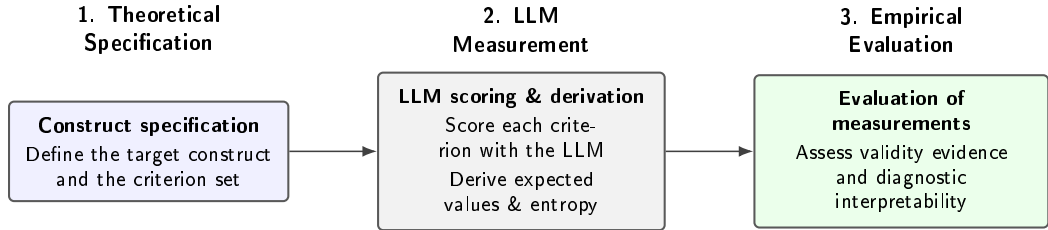
Research questions

Evaluating construct specification

- RQ1** Does explicit formative recomposition of theoretically specified criteria provide stronger validity evidence than holistic or weakly specified alternatives?
- RQ2** Do separately elicited formative criteria support more diagnostically interpretable artefact-level measurements than holistic or weakly specified alternatives?

Measurement procedure

From construct specification to validity evidence



Key idea

The procedure turns a target construct into criterion-level LLM measurements, then evaluates whether recomposed scores provide validity evidence and diagnostic interpretability.

Validation Settings & Experimental Conditions

Testing LLM measurements against external benchmarks

FeedbackQA

Public Q&A dataset

Benchmark: **human ratings**

Specifications compared:

- Naive holistic (1 score)
- Informed holistic (1 score)
- Formative criteria (4 scores)

Barter Deals

Proprietary marketplace data

Benchmark: **structural market baseline**

Outcome: application counts

Specifications compared:

- Naive holistic (1 score)
- Macro-formative (3 scores)
- Fine-grained formative (10 scores)

Key comparison

Do explicit formative LLM measurements provide stronger validity evidence than weaker construct specifications or non-LLM baselines?

Main results

Fine-grained measurement helps most when the construct is separable

FeedbackQA

Convergent validity

- Formative $>$ holistic baselines
- Best Spearman's ρ : **0.578**
- Criteria are strongly intercorrelated

Interpretation:

Decomposition improves alignment with human ratings, but dimensions are not cleanly separable.

Barter Deals

Predictive validity

- Fine-grained formative scores add strong incremental fit
- Best ΔBIC : **+352**
- Poisson deviance: **14.68 $\rightarrow$ 13.23**

Interpretation:

Fine-grained criteria capture meaningful variation in creator application behavior and are cleanly separable.

Conclusion

Moving from holistic prediction to meaningful measurement

The Core Takeaway

Formative LLM measurement provides stronger validity evidence than holistic baselines, but its utility is construct-dependent: it works best when the dimensions are empirically separable.

Broader Implications for the Field:

- **Epistemic shift:** LLM evaluation should be treated as formal measurement, not merely text prediction.
- **Validity framing:** Human ratings and behavioral outcomes provide validity evidence, not infallible “true scores.”
- **Practical design:** Explicit, fine-grained criteria improve both statistical validity and diagnostic transparency.

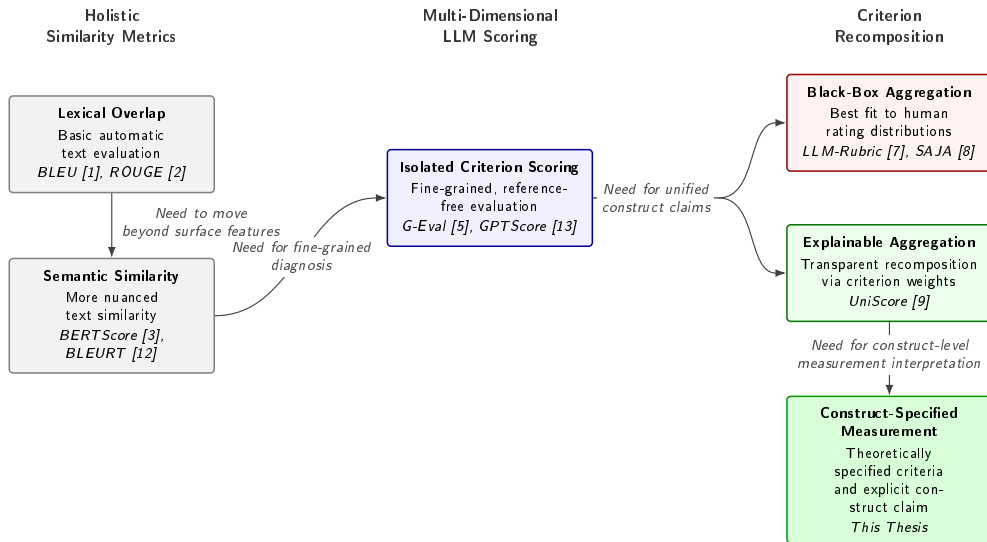
References I

- [1] Kishore Papineni. Bleu: a method for automatic evaluation of mt. *Research Report, Computer Science RC22176 (W0109-022)*, 2001.
- [2] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.
- [3] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*, 2019.
- [4] Wei Zhao, Maxime Peyrard, Fei Liu, Yang Gao, Christian M Meyer, and Steffen Eger. Moverscore: Text generation evaluating with contextualized embeddings and earth mover distance. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 563–578, 2019.
- [5] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: NLG evaluation using GPT-4 with better human alignment. URL <http://arxiv.org/abs/2303.16634>.
- [6] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, E. Xing, Haotong Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. URL <https://www.semanticscholar.org/paper/Judging-LLM-as-a-judge-with-MT-Bench-and-Chatbot-Zheng-Chiang/a0a79dad89857a96f8f71b14238e5237cbfc4787>.

References II

- [7] Helia Hashemi, Jason Eisner, Corby Rosset, Benjamin Van Durme, and Chris Kedzie. LLM-rubric: A multidimensional, calibrated approach to automated evaluation of natural language texts. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13806–13834. doi: 10.18653/v1/2024.acl-long.745. URL <http://arxiv.org/abs/2501.00274>.
- [8] Sneha Kola, Pankaj Kumar Sharma, Soumyadeep Dey, Bamdev Mishra, and Mayur Datar. SAJA: A Simple Approach to Judge Alignment for LLM-as-a-Judge. In *The 64th Annual Meeting of the Association for Computational Linguistics – Industry Track*, April 2026.
- [9] Geonwoo Bang, Dongho Kim, and Moohong Min. Transforming user defined criteria into explainable indicators with an integrated LLM AHP system. URL <http://arxiv.org/abs/2601.05267>.
- [10] Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. SummEval: Re-evaluating summarization evaluation. URL <http://arxiv.org/abs/2007.12626>.
- [11] Kenneth Bollen and Richard Lennox. Conventional wisdom on measurement: A structural equation perspective. *Psychological bulletin*, 110(2):305, 1991.
- [12] Thibault Sellam, Dipanjan Das, and Ankur Parikh. BLEURT: Learning robust metrics for text generation. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7881–7892. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.704. URL <https://aclanthology.org/2020.acl-main.704/>.
- [13] Jinlan Fu, See-Kiong Ng, Zhengbao Jiang, and Pengfei Liu. GPTScore: Evaluate as You Desire, February 2023.

Appendix: Literature lineage



Appendix: Full measurement framework

